

Exercise 1.2

$$\text{let } Z_1 = 3 - 4i, Z_2 = 4 + 5i$$

$$\text{and } Z_3 = -5 + 6i$$

Verify the following

(i) Addition of complex numbers is commutative.

$$Z_1 + Z_2 = Z_2 + Z_1$$

$$(3 - 4i) + (4 + 5i) = (4 + 5i) + (3 - 4i)$$

$$\Rightarrow (3+4) + (-4+5)i = (4+3) + (5-4)i$$

$$\Rightarrow 7 + i = 7 + i$$

(ii) Multiplication of complex numbers is commutative

$$Z_1 Z_2 = Z_2 Z_1$$

$$\Rightarrow (3 - 4i) \cdot (4 + 5i) = (4 + 5i) \cdot (3 - 4i)$$

$$\Rightarrow 12 + 15i - 16i - 20i^2 = 12 - 16i + 15i - 20i^2$$

$$\Rightarrow 12 - i + 20 = 12 - i + 20$$

$$\Rightarrow 32 - i = 32 - i$$

(iii) Addition of complex numbers is associative

$$(Z_1 + Z_2) + Z_3 = Z_1 + (Z_2 + Z_3)$$

$$\Rightarrow (3 - 4i) + (4 + 5i) + (-5 + 6i) = (3 - 4i) + ((4 + 5i) + (-5 + 6i))$$

$$\Rightarrow (7 + i) + (-5 + 6i) = (3 - 4i) + (-1 + 11i)$$

$$\Rightarrow 2 + 7i = 2 + 7i$$

(iv) Multiplication of complex numbers is associative.

$$(Z_1 Z_2) Z_3 = Z_1 (Z_2 Z_3)$$

L.H.S

$$(Z_1 Z_2) Z_3$$

$$\Rightarrow ((3 - 4i) \cdot (4 + 5i)) \cdot (-5 + 6i)$$

$$= (12 + 15i - 16i - 20i^2) \cdot (-5 + 6i)$$

$$= (12 - i + 20) \cdot (-5 + 6i)$$

$$\begin{aligned} &= (32 - i) \cdot (-5 + 6i) \\ &= -160 + 192i + 5i - 6i^2 \\ &= -160 + 6 + 197i \\ &= -154 + 197i \end{aligned}$$

$$\text{R.H.S } Z_1 (Z_2 Z_3)$$

$$= (3 - 4i) \cdot ((4 + 5i) \cdot (-5 + 6i))$$

$$= (3 - 4i) \cdot (-20 + 24i - 25i + 30i^2)$$

$$= (3 - 4i) \cdot (-50 - i)$$

$$= -150 - 3i + 20i + 4i^2$$

$$= -154 + 197i$$

$$\text{R.H.S} = \text{L.H.S}$$

(v) Multiplication of complex numbers is distributive over addition.

$$Z_1 (Z_2 + Z_3) = Z_1 Z_2 + Z_1 Z_3$$

L.H.S

$$Z_1 (Z_2 + Z_3)$$

$$= (3 - 4i) \cdot ((4 + 5i) + (-5 + 6i))$$

$$= (3 - 4i) \cdot (-1 + 11i)$$

$$= -3 + 33i + 4i - 44i^2$$

$$= -3 + 44 + 37i$$

$$= 41 + 37i$$

R.H.S

$$\text{R.H.S } Z_1 Z_2 + Z_1 Z_3$$

$$= (3 - 4i) \cdot (4 + 5i) + (3 - 4i) \cdot (-5 + 6i)$$

$$= (12 + 15i - 16i - 20i^2) + (-15 + 18i + 20i - 24i^2)$$

$$= (32 - i) + (9 + 38i)$$

$$= 41 + 37i$$

Your Success is our win

Adnan ullokh Khan

Q#2 if z_1 and z_2 are complex numbers then prove that:

- $z + \bar{z}$ is real
- $z - \bar{z}$ is imaginary
- $|z_1 \cdot z_2| = |z_1| |z_2|$

(a) $z = a + ib$
 $\bar{z} = a - ib$

$$z + \bar{z} = (a + ib) + (a - ib)$$

$$= a + a + ib - ib$$

$$z + \bar{z} = 2a + 0i \text{ real}$$

$$= 2a$$

(b) $z - \bar{z} = (a + ib) - (a - ib)$

$$= (a - a) + (bi + bi)$$

$$= 0 + 2bi \text{ Im}$$

(c) $|z_1 \cdot z_2| = |z_1| |z_2|$

L.H.S

$$z_1 \cdot z_2 = (a_1 + bi) \cdot (a_2 + bi)$$

$$z_1 \cdot z_2 = a_1 a_2 + a_1 b_2 i + a_2 b_1 i + b_1 b_2 i^2$$

$$z_1 \cdot z_2 = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$$

$$|z_1 \cdot z_2| = \sqrt{(a_1 a_2 - b_1 b_2)^2 + (a_1 b_2 + a_2 b_1)^2}$$

$$|z_1 \cdot z_2| = \sqrt{(a_1 a_2)^2 + 2a_1 a_2 b_1 b_2 + (b_1 b_2)^2 + (a_1 b_2)^2 + 2a_1 b_2 a_2 b_1 + (a_2 b_1)^2}$$

$$|z_1 \cdot z_2| = \sqrt{(a_1 a_2)^2 + (b_1 b_2)^2 + (a_1 b_2)^2 + (a_2 b_1)^2}$$

R.H.S $|z_1| |z_2|$

$$z_1 = a_1 + ib_1$$

$$|z_1| = \sqrt{a_1^2 + b_1^2}$$

$$z_2 = a_2 + ib_2$$

$$|z_2| = \sqrt{a_2^2 + b_2^2}$$

Adnan
Ullah
Khan

$$|z_1| |z_2| = \sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2} \quad (7)$$

$$|z_1| |z_2| = \sqrt{(a_1^2 + b_1^2) \cdot (a_2^2 + b_2^2)}$$

$$|z_1| |z_2| = \sqrt{a_1^2 a_2^2 + a_1^2 b_2^2 + b_1^2 a_2^2 + b_1^2 b_2^2}$$

$$|z_1| |z_2| = \sqrt{(a_1 a_2)^2 + (b_1 b_2)^2 + (a_1 b_2)^2 + (a_2 b_1)^2}$$

R. L.H.S = R.H.S

Q#3 Find the additive inverses of the following complex numbers

(i) $3 + 5i$

$$-z = -(3 + 5i) = -3 - 5i$$

(ii) $6 - 5i$

Additive Inverse $-6 + 5i$

(iii) $(11, 0)$

Additive Inverse $(-11, 0)$

(iv) $(\frac{2}{3}, 6)$

Additive Inverse $(-\frac{2}{3}, -6)$

(v) $(\frac{8}{9}, -\frac{4}{5})$

Additive Inverse $(-\frac{8}{9}, \frac{4}{5})$

(vi) $(0, \frac{3}{8})$

Additive Inverse $(0, -\frac{3}{8})$

Q#4 Find the multiplicative inverses of the following complex numbers.

(i) $3 + 5i$

$$z = 3 + 5i$$

$$\frac{1}{z} = \frac{1}{3 + 5i}$$

$$\frac{1}{z} = \frac{1}{3 + 5i} \cdot \frac{3 - 5i}{3 - 5i}$$

$$\Rightarrow \frac{1}{z} = \frac{3 - 5i}{3^2 - (5i)^2} = \frac{3 - 5i}{9 + 25} = \frac{3 - 5i}{34}$$

Q#4 ii) $8i$

$$z = 0 + 8i$$

$$\Rightarrow \frac{1}{z} = \frac{1}{0+8i}$$

$$\Rightarrow \frac{1}{z} = \frac{1}{0+8i} \times \frac{0-8i}{0-8i} = \frac{-8i}{-64i^2}$$

$$\Rightarrow \frac{1}{z} = 0 + \frac{i}{8}$$

iii) $(10, 4)$

$$z = (a, b), \quad \bar{z}^{-1} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2} \right)$$

$$\bar{z}^{-1} = \left(\frac{10}{10^2+4^2}, \frac{-4}{10^2+4^2} \right) = \left(\frac{10}{116}, \frac{-4}{116} \right)$$

$$\bar{z}^{-1} = \left(\frac{2}{58}, \frac{-1}{29} \right)$$

iv) $(12, -7)$

$$\bar{z}^{-1} = \left(\frac{12}{12^2+7^2}, \frac{-(-7)}{12^2+7^2} \right)$$

$$= \left(\frac{12}{144+49}, \frac{7}{144+49} \right)$$

$$= \left(\frac{12}{193}, \frac{7}{193} \right)$$

v) $(-8, 0)$

$$\bar{z}^{-1} = \left(\frac{-8}{(-8)^2+0^2}, \frac{0}{(-8)^2+0^2} \right)$$

$$\bar{z}^{-1} = \left(\frac{1}{8}, 0 \right)$$

vi) $(0, -\frac{3}{4})$

$$= \left(\frac{0}{0^2+(-\frac{3}{4})^2}, \frac{-(-\frac{3}{4})}{0^2+(-\frac{3}{4})^2} \right)$$

$$= \left(0, \frac{1}{\frac{3}{4}} \right) = \left(0, \frac{4}{3} \right)$$

Q#5 Find The additive ⑧ and multiplicative Inverse by definition

i) $(2, 3)$ ii) $(-4, 5)$

Additive Inverse

$$z + (-z) = 0$$

$$\Rightarrow (a, b) + (c, d) = 0$$

$$\Rightarrow (2, 3) + (c, d) = 0$$

$$2+c=0, \quad 3+d=0$$

$$c = -2, \quad d = -3$$

$$-z = (-2, -3)$$

Multiplicative Inverse

$$z \bar{z}^{-1} = 1$$

$$\Rightarrow (a, b) \times (c, d) = (1, 0)$$

$$\Rightarrow (2, 3) \times (c, d) = (1, 0)$$

$$\Rightarrow (2c-3d, 2d+3c) = (1, 0)$$

$$\Rightarrow 2c-3d = 1 \quad \text{--- (1)}$$

$$\Rightarrow 2d+3c = 0 \quad \text{--- (2)}$$

$$\Rightarrow d = \frac{-3c}{2} \text{ Put in (1)}$$

$$\Rightarrow 2c - 3\left(\frac{-3c}{2}\right) = 1$$

$$\Rightarrow 2c + \frac{9c}{2} = 1$$

$$\Rightarrow \frac{4c+9c}{2} = 1$$

$$\Rightarrow \frac{13c}{2} = 1$$

$$\Rightarrow c = \frac{2}{13}$$

$$d = \frac{-3c}{2} = \frac{-3\left(\frac{2}{13}\right)}{2} = \frac{-3}{13}$$

$$\bar{z}^{-1} = \left(\frac{2}{13}, \frac{-3}{13} \right)$$

YOUR
Success
is
OUR
Win

(ii) $(-4, 5)$

Additive Inverse

$$z + (-z) = 0$$

$$\Rightarrow (a, b) + (c, d) = (0, 0)$$

$$\Rightarrow (-4, 5) + (c, d) = (0, 0)$$

$$\Rightarrow (-4 + c, 5 + d) = (0, 0)$$

$$\Rightarrow (-4 + c, 5 + d) = (0, 0)$$

$$-4 + c = 0, \quad 5 + d = 0$$

$$c = 4 \quad d = -5$$

$$-z = (4, -5)$$

Multiplicative Inverse

$$z z^{-1} = 1$$

$$(a, b) \cdot (c, d) = (1, 0)$$

$$\Rightarrow (ac - bd, ad + bc) = (1, 0)$$

$$\Rightarrow (-4, 5) \cdot (c, d) = (1, 0)$$

$$\Rightarrow (-4c - 5d, -4d + 5c) = (1, 0)$$

$$\Rightarrow -4c - 5d = 1, \quad -4d + 5c = 0$$

$$\Rightarrow 5c = 4d$$

$$\Rightarrow c = \frac{4}{5}d$$

$$\Rightarrow -4\left(\frac{4}{5}d\right) - 5d = 1$$

$$\Rightarrow -\frac{16}{5}d - 5d = 1$$

$$\Rightarrow \frac{-16 - 25d}{5} = 1$$

$$\Rightarrow -\frac{41d}{5} = 1$$

$$\Rightarrow d = -\frac{5}{41}$$

$$c = \frac{4}{5}d = \frac{4}{5}\left(-\frac{5}{41}\right) = -\frac{4}{41}$$

$$z^{-1} = \left(-\frac{4}{41}, -\frac{5}{41}\right)$$

Q#6. Verify the following Properties with

$$z = 4 - 3i, \quad z_1 = 3 - 2i \text{ and } z_2 = 2 + 3i$$

(i) $|z| = |-z| = |\bar{z}| = |- \bar{z}|$

For $|z|$

$$z = 4 - 3i$$

$$|z| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = 5$$

For

$$-z = -4 + 3i$$

$$|-z| = \sqrt{(-4)^2 + (3)^2} = \sqrt{16 + 9} = 5$$

For $|\bar{z}|$

$$\bar{z} = 4 + 3i$$

$$|\bar{z}| = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = 5$$

For $|- \bar{z}|$

$$z = 4 + 3i$$

$$-\bar{z} = -(4 + 3i)$$

$$-\bar{z} = -4 - 3i$$

$$|- \bar{z}| = \sqrt{(-4)^2 + (-3)^2} = \sqrt{16 + 9}$$

$$|- \bar{z}| = \sqrt{25} = 5$$

(ii) $\bar{\bar{z}} = z$

$$z = 4 - 3i$$

$$\bar{z} = 4 + 3i$$

$$\bar{\bar{z}} = 4 - 3i = z$$

(iii) $z \bar{z} = |z|^2$

L.H.S

$$z = 4 - 3i$$

$$\bar{z} = 4 + 3i$$

$$z \bar{z} = (4 - 3i)(4 + 3i)$$

$$z \bar{z} = (4)^2 - (3i)^2 = 16 + 9 = 25$$

R.H.S $|z| = \sqrt{4^2 + 3^2} = \sqrt{25}$

$$|z|^2 = 25 \quad \text{L.H.S} = \text{R.H.S}$$

Adnan
Ullah
Khan

$$(10) \overline{Z_1 + Z_2} = \overline{Z_1} + \overline{Z_2}$$

L.H.S

$$Z_1 + Z_2 = (3 - 2j) + (2 + 3j)$$

$$Z_1 + Z_2 = (5 + j)$$

$$\overline{Z_1 + Z_2} = 5 - j$$

R.H.S

$$Z_1 = 3 - 2j$$

$$\overline{Z_1} = 3 + 2j$$

$$Z_2 = 2 + 3j$$

$$\overline{Z_2} = 2 - 3j$$

$$\overline{Z_1} + \overline{Z_2} = (3 + 2j) + (2 - 3j)$$

$$\overline{Z_1} + \overline{Z_2} = (5 - j)$$

L.H.S = R.H.S

$$(11) \overline{Z_1 Z_2} = \overline{Z_1} \cdot \overline{Z_2}$$

L.H.S

$$Z_1 \cdot Z_2 = (3 - 2j) \cdot (2 + 3j)$$

$$\Rightarrow Z_1 \cdot Z_2 = 6 + 9j - 4j - 6j^2$$

$$\Rightarrow Z_1 \cdot Z_2 = 12 + 5j$$

$$\Rightarrow \overline{Z_1 \cdot Z_2} = 12 - 5j$$

R.H.S

$$\overline{Z_1} \cdot \overline{Z_2} = (3 + 2j)(2 - 3j)$$

$$\Rightarrow \overline{Z_1} \cdot \overline{Z_2} = 6 - 9j + 4j - 6j^2$$

$$\Rightarrow \overline{Z_1} \cdot \overline{Z_2} = 12 - 5j$$

$$(12) \left(\frac{Z_1}{Z_2} \right) = \frac{\overline{Z_1}}{\overline{Z_2}}$$

L.H.S

$$\frac{Z_1}{Z_2} = \frac{3 - 2j}{2 + 3j}$$

$$= \frac{3 - 2j}{2 + 3j} \times \frac{2 - 3j}{2 - 3j}$$

$$\frac{Z_1}{Z_2} = \frac{6 - 9j - 4j + 6j^2}{4 + 9}$$

$$\frac{Z_1}{Z_2} = \frac{0 + 13j}{13} = 0 + j$$

$$\left(\frac{Z_1}{Z_2} \right) = 0 + j$$

R.H.S

$$\frac{\overline{Z_1}}{\overline{Z_2}} = \frac{3 + 2j}{2 - 3j}$$

$$\Rightarrow \frac{\overline{Z_1}}{\overline{Z_2}} = \frac{3 + 2j}{2 - 3j} \times \frac{2 + 3j}{2 + 3j}$$

$$= \frac{6 + 9j + 4j + 6j^2}{4 - 9j^2}$$

$$= \frac{0 + 13j}{13} = 0 + j$$

L.H.S = R.H.S

Q#7). if $Z_1 = 3 - 2j$ and $Z_2 = 2 - 3j$, then express the following in the form of $a + jb$

$$(i) \frac{Z_1 \overline{Z_2}}{\overline{Z_1}} = \frac{(3 - 2j) \cdot (2 - 3j)}{(3 - 2j)}$$

$$\frac{Z_1 \overline{Z_2}}{\overline{Z_1}} = \frac{(3 - 2j)(2 - 3j)}{3 - 2j}$$

$$\frac{Z_1 \overline{Z_2}}{\overline{Z_1}} = \frac{12 + 5j}{3 - 2j}$$

$$= \frac{12 + 5j}{3 - 2j} \times \frac{3 + 2j}{3 + 2j}$$

$$= \frac{46 - 9j}{9 + 4} = \frac{46}{13} - \frac{9j}{13}$$

$$(ii) \frac{\overline{Z_1} Z_2}{\overline{Z_2}} = \frac{(3 + 2j)(2 - 3j)}{2 - 3j}$$

$$= \frac{(3 + 2j)(2 - 3j)}{2 - 3j}$$

$$= \frac{12 - 5j}{2 - 3j}$$

$$\begin{aligned}\frac{\bar{z}_1 z_2}{\bar{z}_2} &= \frac{12-5j}{2+3j} \\ &= \frac{12-5j}{2+3j} \times \frac{2-3j}{2-3j} \\ &= \frac{9-46j}{13} = \frac{9}{13} - \frac{46j}{13}\end{aligned}$$

Triangle Inequality

$$|z_1| - |z_2| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

$$z_1 = 3-2j, \quad z_2 = 2-3j$$

$$z_1 + z_2 = 5-5j$$

$$|z_1| = \sqrt{9+4} = \sqrt{13}$$

$$|z_2| = \sqrt{4+9} = \sqrt{13}$$

$$\begin{aligned}|z_1 + z_2| &= \sqrt{5^2 + 5^2} \\ &= \sqrt{25+25}\end{aligned}$$

$$|z_1 + z_2| = 5\sqrt{2}$$

$$|z_1| - |z_2| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

$$\sqrt{13} - \sqrt{13} \leq 5\sqrt{2} \leq \sqrt{13} + \sqrt{13}$$

$$0 \leq 5\sqrt{2} \leq 2\sqrt{13}$$

Proved.

Q#8 Find the real and imaginary parts of each of the following by using any method.

$$\textcircled{i} (\sqrt{2} + j)^{-2}$$

$$= \frac{1}{(\sqrt{2} + j)^2}$$

$$= \frac{1}{2+2\sqrt{2}j+j^2}$$

$$= \frac{1}{1+2\sqrt{2}j} \quad \textcircled{ii}$$

$$= \frac{1}{1+2\sqrt{2}j} \times \frac{1-2\sqrt{2}j}{1-2\sqrt{2}j}$$

$$= \frac{1-2\sqrt{2}j}{1-4(2)j^2}$$

$$= \frac{1-2\sqrt{2}j}{9} = \frac{1}{9} - \frac{2\sqrt{2}j}{9}$$

$$\begin{aligned}\text{Real } \frac{1}{9} \\ \text{Imag } -\frac{2\sqrt{2}}{9}\end{aligned}$$

$$\textcircled{ii} (1-\sqrt{5}j)^{-1}$$

$$= \frac{1}{1-\sqrt{5}j}$$

$$= \frac{1}{1-\sqrt{5}j} \times \frac{1+\sqrt{5}j}{1+\sqrt{5}j}$$

$$= \frac{1+\sqrt{5}j}{1+5} = \frac{1}{6} + \frac{\sqrt{5}j}{6}$$

$$\text{Re} = \frac{1}{6}, \quad \text{Im} = \frac{\sqrt{5}}{6}$$

$$\textcircled{iii} (\sqrt{3}-j)^2$$

$$= \frac{1}{(\sqrt{3}-j)^2}$$

$$= \frac{1}{(\sqrt{3})^2 - 2\sqrt{3}j + j^2}$$

$$= \frac{1}{3-2\sqrt{3}j-1} = \frac{1}{2-2\sqrt{3}j}$$

$$= \frac{1}{2-2\sqrt{3}j} \times \frac{2+2\sqrt{3}j}{2+2\sqrt{3}j}$$

$$= \frac{2+2\sqrt{3}j}{4+(4)(3)} = \frac{2}{16} + \frac{2\sqrt{3}j}{16}$$

$$\text{Re} = \frac{1}{8}, \quad \text{Im} = \frac{\sqrt{3}}{8}$$

$$\begin{aligned}
 \textcircled{iv} \quad (2i - \sqrt{3})^{-1} &= \frac{1}{2i - \sqrt{3}} \times \frac{-2i - \sqrt{3}}{-2i - \sqrt{3}} \\
 &= \frac{-2i - \sqrt{3}}{(2i - \sqrt{3})(-2i - \sqrt{3})} \\
 &= \frac{-\sqrt{3} - 2i}{(-\sqrt{3} + 2i)(-\sqrt{3} - 2i)} \\
 &= \frac{-\sqrt{3} - 2i}{(-\sqrt{3})^2 - (2i)^2} \\
 &= \frac{-\sqrt{3} - 2i}{3 + 4} \\
 &= \frac{-\sqrt{3}}{7} - \frac{2}{7}i \\
 \text{Re} &= -\frac{\sqrt{3}}{7} \\
 \text{Im} &= -\frac{2}{7}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{v} \quad \left(\frac{1}{4i-5}\right)^{-1} &= \frac{1}{(4i-5)^{-1}} \\
 &= 4i-5 \\
 &= -5+4i \\
 \text{Re} &= -5 \quad \text{Im} = +4
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{vi} \quad \left(\frac{3+4i}{5i-4}\right)^2 &= \left(\frac{5i-4}{3+4i}\right)^2 \\
 &= \frac{(-4+5i)^2}{(3+4i)^2}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{16 + 2(-4)(5i) + (5i)^2}{9 + 24i + i^2(16)} \quad \textcircled{12} \\
 &= \frac{16 - 25 - 40i}{-7 + 24i} \\
 &= \frac{-9 - 40i}{-7 + 24i} = \frac{-9 - 40i}{-7 + 24i} \\
 &= \frac{-9 - 40i}{-7 + 24i} \times \frac{-7 - 24i}{-7 - 24i} \\
 &= \frac{-897 + 496i}{625} \\
 &= -\frac{897}{625} + \frac{496}{625}i
 \end{aligned}$$

$$\text{Re} = -\frac{897}{625}, \quad \text{Im} = \frac{496}{625}$$

$$\begin{aligned}
 \textcircled{vii} \quad \left(\frac{3i-2}{2-3i}\right)^{-2} &= \left(\frac{2-3i}{3i-2}\right)^2 \\
 &= \left(\frac{2-3i}{-2+3i}\right)^2 \\
 &= \left(\frac{2-3i}{-(2-3i)}\right)^2 \\
 &= \left(\frac{1}{-1}\right)^2 = 1 + 0i \\
 \text{Re} &= 1 \quad \text{Im} = 0
 \end{aligned}$$

Adnan Ulh Khan
Your Success is
our win